MTH 201 Skills Check 1 – Practice

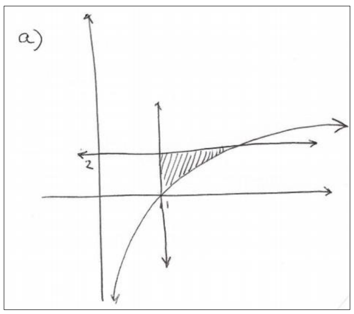
This skills check covers pre-requisite material needed for the first unit of the course, areas and volumes

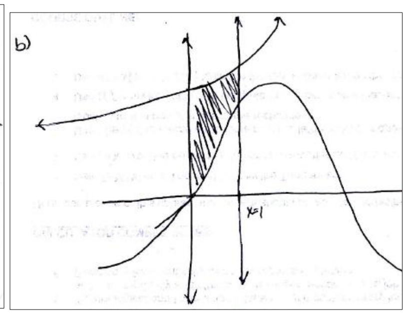
1. **Identifying Bounded Regions**

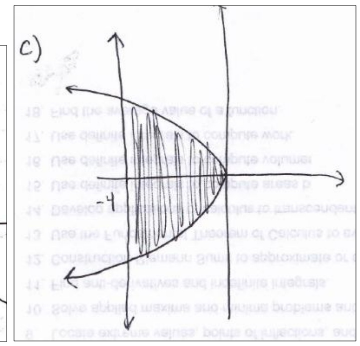
a) Shade the region bounded by the following curves: $h(x) = \ln(x),\ x = 1,\ y = 2$

b) Shade the region bounded by the following curves: $f(x) = \sin(x),\ g(x) = e^x,\ x = 0,\ x = 1$

c) Shade the region bounded by the following curves: $x = -y^2,\ x = -4$







2. **Finding Points of Intersection**

a) Find the points of intersection of the curves: $y = x^2 + x \quad \& \quad y = 2x + 6$

$$
\begin{aligned}
& x^2 + x = 2x + 6 \\
\Leftrightarrow\ & x^2 - x - 6 = 0 \\
\Leftrightarrow\ & (x-3)(x+2) = 0 \\
\Leftrightarrow\ & x = 3 \ or \ x = -2
\end{aligned}
$$

Therefore the *points* of intersection are:

$$(3,12) \ \& \ (-2,2)$$

b) Find the points of intersection of the curves: $f(x)=\ln(2x)$ & $g(x)=0$

$$\begin{aligned} &\ln(2x)=0 \\ \Leftrightarrow\ & e^{\ln(2x)}=e^{0} \\ \Leftrightarrow\ & 2x=1 \\ \Leftrightarrow\ & x=\frac{1}{2} \end{aligned}$$

Therefore the *point* of intersection is:

$$\left(\frac{1}{2},0\right)$$

c) Find the points of intersection of the curves: $y=\sqrt{2x+7}$ & $y=x+2$

$$\begin{aligned} &\sqrt{2x+7}=x+2 \\ \Leftrightarrow\ & 2x+7=(x+2)^2 \\ \Leftrightarrow\ & 2x+7=x^2+4x+4 \\ \Leftrightarrow\ & x^2+2x-3=0 \\ \Leftrightarrow\ & (x+3)(x-1)=0 \\ \Leftrightarrow\ & x=-3 \text{ or } x=1 \end{aligned}$$

Since both sides of the equations were squared during the solving process we must check for extraneous solutions. If we check both potential solutions we find that $x=-3$ is an extraneous solution!

Therefore the only *point* of intersection is:

$$(1,3)$$

3. Definite Integrals for Rotations

a) Evaluate the definite integral: $\int_0^3 (x+2)^2\,dx$

$$\int_0^3 (x+2)^2\,dx=\int_0^3 \left(x^2+4x+4\right)dx=\left(\frac{1}{3}x^3+2x^2+4x\right)\Big|_0^3=\left(\frac{1}{3}(27)+2(9)+12\right)-0=39$$

b) Evaluate the definite integral: $\int_1^0 \left(\pi e^t - 1\right) dt$

$$\int_1^0 \left(\pi e^t - 1\right) dt = \left(\pi e^t - t\right)\Big|_1^0 = \left(\pi e^0 - 0\right) - \left(\pi e - 1\right) = \pi - \pi e + 1$$

c) Evaluate the definite integral: $\int_1^2 \pi \left(x^2 - \frac{1}{x}\right) dx$

$$\int_1^2 \pi \left(x^2 - \frac{1}{x}\right) dx = \pi \left(\frac{1}{3}x^3 - \ln|x|\right)\Big|_1^2 = \pi\left(\frac{8}{3} - \ln(2)\right) - \pi\left(\frac{1}{3} - \ln(1)\right) = \frac{7\pi}{3} - \pi \ln(2)$$

4. U-Substitutions with Definite Integrals

a) Use a u-substitution to evaluate the definite integral: $\int_0^{\pi/4} \frac{\sin(x)}{\cos(x)} dx$

Let $u = \cos(x)$ and thus $du = -\sin(x)dx \quad \Leftrightarrow \quad -du = \sin(x)dx$.

Our limits of integration become: $u(0) = \cos(0) = 1$ & $u\left(\frac{\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$.

$$\int_0^{\pi/4} \frac{\sin(x)}{\cos(x)} dx = \int_1^{\sqrt{2}/2} -\frac{1}{u} du = \left(-\ln|u|\right)\Big|_1^{\sqrt{2}/2} = -\ln\left(\frac{\sqrt{2}}{2}\right) + \ln(1) = -\ln\left(\frac{\sqrt{2}}{2}\right)$$

b) Use a u-substitution to evaluate the definite integral: $\int_0^1 \frac{e^x}{e^x + 1} dx$

Let $u = e^x + 1$ and thus $du = e^x dx$.
Our limits of integration become: $u(0) = e^0 + 1 = 2$ & $u(1) = e + 1$.

$$\int_0^1 \frac{e^x}{e^x + 1} dx = \int_2^{e+1} \frac{1}{u} du = \ln|u|\Big|_2^{e+1} = \ln(e+1) - \ln(2) = \ln\left(\frac{e+1}{2}\right)$$

c) Use a u-substitution to evaluate the definite integral: $\int_1^2 x\sqrt{x^2-1}\,dx$

Let $u=x^2-1$ and thus $du=2x\,dx \;\Leftrightarrow\; \frac{1}{2}du=x\,dx$.

Our limits of integration become: $u(1)=(1)^2-1=0$ & $u(2)=(2)^2-1=3$

$$\int_1^2 x\sqrt{x^2-1}\,dx=\int_0^3 \frac{1}{2}\sqrt{u}\,du=\frac{1}{2}\int_0^3 u^{\frac{1}{2}}\,du=\frac{1}{2}\left(\frac{2}{3}u^{\frac{3}{2}}\right)\Bigg|_0^3=\frac{1}{3}(3)^{\frac{3}{2}}-\frac{1}{3}(0)^{\frac{3}{2}}=\frac{1}{3}\sqrt{27}-0=\sqrt{3}$$

5. Solving Literal Equations

a) Solve the equation for *x*: $y=\frac{2}{x+3}$

$$\begin{aligned} &y(x+3)=2\\ \Leftrightarrow\;& x+3=\frac{2}{y}\\ \Leftrightarrow\;& x=\frac{2}{y}-3 \end{aligned}$$

b) Solve the equation for *x*: $y=2\sqrt[5]{x+1}$

$$\begin{aligned} &y=2\sqrt[5]{x+1}\\ \Leftrightarrow\;& \frac{y}{2}=\sqrt[5]{x+1}\\ \Leftrightarrow\;& \left(\frac{y}{2}\right)^5=x+1\\ \Leftrightarrow\;& \frac{y^5}{32}-1=x \end{aligned}$$

c) Solve the equation for *x*: $y=\ln(x+2)$

$$\begin{aligned} &y=\ln(x+2)\\ \Leftrightarrow\;& e^y=e^{\ln(x+2)}\\ \Leftrightarrow\;& e^y=x+2\\ \Leftrightarrow\;& e^y-2=x \end{aligned}$$